

Tractable Problems in *AI Security* via Formal Methods

The one-sentence pitch

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Or: take a requirements list off the document and pay them to build the widget.

What formal methods can and can't claim

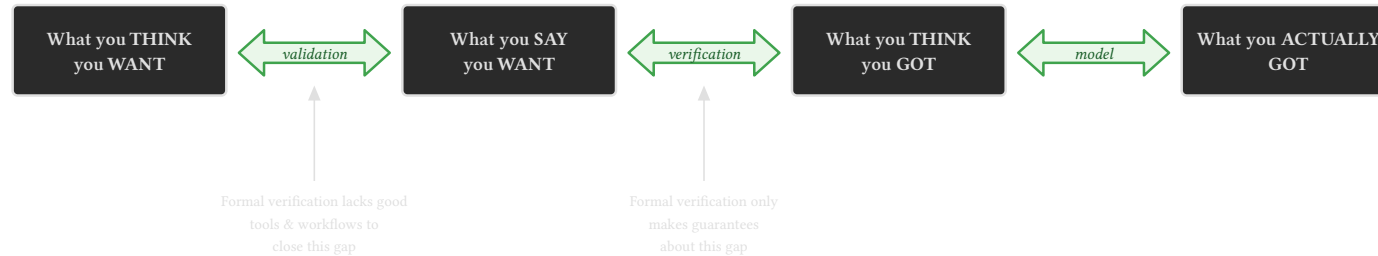


Figure 1: After Evan Miyazono. FM closes the middle gap only.

What FM can and can't claim, in words

- **Elicitation gap** (left): is the spec the thing we actually wanted? Out of scope for the proof.
- **Verification gap** (middle): does the implementation meet the spec? This is what FM does.
- **Modeling gap** (right): does the proof's model of the hardware/OS/network match reality?

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Most real-world failures live on the ends. A widget without an honest story for both ends is not a widget.

The ML stack, in five layers

1. **Execution harness** — inference servers, sandboxes, API gateways
2. **Software & ML framework** — compilers, PyTorch/JAX, dependency supply chain
3. **Orchestration & cloud** — K8s/Slurm/Ray, fabric, IAM
4. **Firmware & low-level** — hypervisors, device drivers, boot integrity
5. **Hardware & physical** — RTL, secure enclaves, side channels

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Each layer has a status quo and an attack surface. Each layer has FM-shaped widgets that would close part of it.

Stack: who's attacking what

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- A **co-tenant agent** breaks containment at L1 or L4.
- A **malicious model** with tool use breaks the sandbox at L1, the runtime at L3.
- A **supply chain** compromise lands in the loader at L2 or kernel binaries at L4.
- A **physical adversary** or insider goes after the fabric at L3 or the chip at L5.

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The widgets pair off against these threats. The matrix on the website is the many-to-many of layers × problems × adversaries.

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- **Enablers** unbottleneck a class of downstream work. Examples: adversarial robustness of FM tools, neuralese governance, spec elicitation.
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An enabler with many dependents should be addressed early. A widget contingent on an unsolved enabler should be sequenced accordingly.

Deep dive: five widgets

I'll spend a slide or two on each.

1. Verified network tap
2. OCI runtime hardening
3. seL4 native GPU drivers
4. AI control via proof-carrying code
5. Adversarial robustness of FM

Widget 1: Verified network tap

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Costs (Cankaya et al., 2026): under 0.01% of facility upfront for the edge tap; 0.3–0.75% for east-west. FPGA capture at 400 Gbps is a solved hardware problem (Arista 7130, Deutsche Börse).

Verified network tap: the LTL contract

$\Box(\text{live_in}(p) \rightarrow \Diamond_{\leq D} \text{live_out}(p))$	(P1) forwarding completeness
$\Box(\text{live_out}(p) \rightarrow \blacklozenge \text{live_in}(p))$	(P2) forwarding fidelity
$\Box(\text{live_in}(p) \rightarrow \Diamond_{\leq D'} \text{mon}(\text{image}(p)))$	(P3) mirror completeness
$\Box(\text{mon}(r) \rightarrow \blacklozenge(\exists p. \text{live_in}(p) \wedge \text{image}(p) = r))$	(P4) mirror soundness
$\Box(\text{live_out}(p) \rightarrow \neg \text{depends-on}(p, \text{mon-input}))$	(P5) no back-channel
$\Box(\text{fail} \rightarrow \Box(\text{live_in}(p) \rightarrow \Diamond_{\leq D} \text{live_out}(p)))$	(P6) fail-to-wire

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P5 is the noninterference property that defeats verifier collusion. P6 is fail-open: a fault must downgrade to a transparent pass-through, never a link cut.

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When the workload is an agent with tool use

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Empirical work: BoxArena (Dougherty 2026) inverts the offensive framing. Fix the attacker model, vary the runtime. Five attack surfaces (fs, socket, process, network, syscall) across runc, runsc, crun, Kata.

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- A formal model of the Linux ABI surface a hardened container actually uses (~80 syscalls, not 350+).
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Lineage: CLInc (Bevier 1989, gates to apps), seL4 (Klein 2009, microkernel functional correctness). Same shape, different target.

Widget 3: seL4 native GPU drivers

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The methodology (separation logic, abstract HW model) is settled. The blocker is that **the abstract hardware model does not exist**. No GPU vendor publishes a machine-checkable spec of their command processor.

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Two stopping points:

- Verified module against a stubbed hardware model — shippable as reference.
- Same driver against a model co-developed with the vendor or with REMS (Sewell) — the research contribution.

Widget 4: AI control via proof-carrying code

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- **PCC** (Necula 1997): every action arrives with a machine-checkable certificate.
Trusted base = proof kernel.
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Midpoint: untrusted agent emits code *and a proof*. The boundary rejects anything that does not typecheck. Trusted base = kernel + human-written spec.

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Three subproblems determine where k lands:

1. Proofs as a matter of degree (graded gatekeepers, not pass/fail).
2. Specialist blue-team models for narrow Hoare fragments.
3. Who pays for tooling — red team's compliance budget, or blue team's?

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This is the enabler with the most dependents. A scheming model that finds a kernel zero-day defeats the entire stack of widgets in one shot. It also gates AI-control-via-PCC

Widget 5 (enabler): Adversarial robustness of FM

the trusted base in that protocol *is* the kernel.

Adversarial robustness: the sketch

Soundness bugs are downstream of the complexity of dependent type theory. The proof checker (kernel) is large.

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Also a governance question: who is the maintainer when the bug is found by an agent at 3am?

Why this is hard to fund

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FM talent today is concentrated in avionics, defense, automotive

Why this is hard to fund

domains where regulators force the comparison to be against a counterfactual incident, not against the free alternative.

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That is what the document is.

What to do this weekend

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- **Reproduce a BoxArena entry.** Pick a runtime, write the harness, run the five surfaces, publish the leaderboard row.
- **Spec a tiny piece of the OCI ABI in Lean or Rocq.** Even ten syscalls is useful.
- **Toy verified network tap in Cryptol or SymbiYosys.** P1, P2, P6 only. Stub the SerDes.
- **Measure k on one spec shape.** Pick context-window allocation. Pick one model. Publish the distribution.
- **Find a soundness bug in Agda or Lean.** Submit it to the Arena.

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- If you're a **PL/FM researcher** looking for an AI-safety entry point: control protocols as a process calculus, or the verified OCI runtime.
- If you're an **ML person** worried about agentic deployment: BoxArena, sampler verification, context-window integrity.
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The website is the live menu: tractable.for-all.dev. The tags are layer × adversary × category. Filter to your shape.

Summary

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Questions?

tractable.for-all.dev
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Backup: the full problem list

Enablers: adversarial robustness of FM, spec elicitation, neuralese governance, AI control via PCC.

Widgets, by layer:

- L1 (harness): verified input parsers, sampler verification, context-window integrity, capability accumulation, agent weird machines, loop termination, audit log integrity.
- L2 (framework): weight integrity (also L1/L4).
- L3 (orchestration): OCI runtime hardening, scheduler co-tenancy, edge policy verification, verified network tap.
- L4 (firmware): seL4 native GPU drivers.
- L5 (hardware): verified network tap (also L3).

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The toolchain is not the bottleneck. The targets are.

Backup: what's out of scope

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- Proving the model itself. Different document.
- Scheduler optimization. Policy, not mechanism.
- Multimode fiber taps. Link budget does not admit passive splitting at high baud rates.
- Full POSIX proofs. API surface too large.
- Anything where the spec cannot factor cleanly from the implementation.